

第十一章 电路的频率响应

- 网络函数
- RLC 串联谐振电路
- RLC 并联谐振电路
- 波特图
- 滤波器简介

§ 11-1 网络函数

- 频率响应：电路或系统的工作状态随频率变化的现象。
- 通常采用单输入-单输出的方式，在输入变量和输出变量之间建立函数关系
- 电路在一个正弦电源激励下达到稳定状态时，各部分的响应都是同频率的正弦量。
- 网络函数的定义：

$$H(j\omega) = \frac{\dot{R}_k(j\omega)}{\dot{E}_{sj}(j\omega)}$$

$\dot{R}_k(j\omega)$ 为输出端口 k 的响应，

$\dot{E}_{sj}(j\omega)$ 为输入端口 j 的输入变量。

§ 11-1 网络函数

$$H(j\omega) = \frac{R_k(j\omega)}{E_{sj}(j\omega)} = A(\omega) \angle \varphi(\omega)$$

电路响应

频率函数

电路激励

$$A(\omega) = \left| \frac{R_k(j\omega)}{E_{sj}(j\omega)} \right|$$

—— 幅频特性

$$\varphi(\omega)$$

—— 相频特性

§ 11-1 网络函数

$$H(j\omega) = \frac{R_k(j\omega)}{E_{sj}(j\omega)} = A(\omega) \angle \varphi(\omega)$$

- 频率响应是指正弦电路的稳态响应与输入信号的频率之间的关系特性。

当 $k=j$ 时，网络函数为一端口的阻抗或导纳；

$$Z = \frac{\dot{U}_k}{\dot{I}_{sk}} \quad Y = \frac{\dot{I}_s}{\dot{U}_{sk}}$$

当 $k \neq j$ 时，网络函数为不同端口的转移函数。

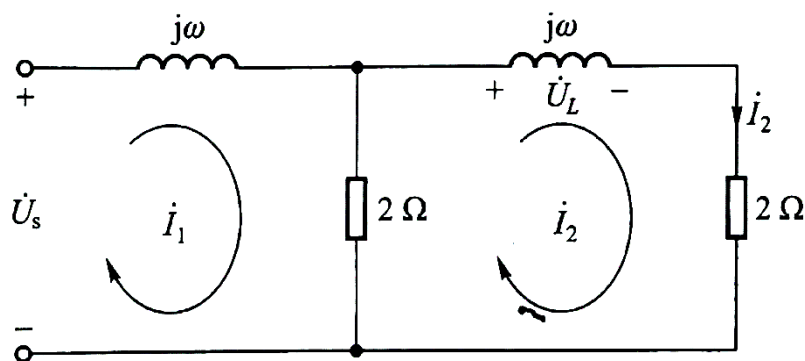
统称为转移函数，分别为：

$$\text{转移阻抗} \quad \frac{\dot{U}_k}{\dot{I}_{sj}} \quad \text{转移电流比} \quad \frac{\dot{I}_k}{\dot{I}_{sj}}$$

$$\text{转移导纳} \quad \frac{\dot{I}_k}{\dot{U}_{sj}} \quad \text{转移电压比} \quad \frac{\dot{U}_k}{\dot{U}_{sj}}$$

例11-1

求图 11-1 所示电路的网络函数 $\frac{\dot{I}_2}{\dot{U}_s}$ 和 $\frac{\dot{U}_L}{\dot{U}_s}$ 。



解 列网孔方程解电流 $\dot{I}_2$

$$\begin{aligned}(2 + j\omega) \dot{I}_1 - 2 \dot{I}_2 &= \dot{U}_s \\ -2 \dot{I}_1 + (4 + j\omega) \dot{I}_2 &= 0\end{aligned}$$

解得电流 $\dot{I}_2$ 为

$$\dot{I}_2 = \frac{2 \dot{U}_s}{4 + (j\omega)^2 + j6\omega}$$

网络函数分别为

$$\frac{\dot{I}_2}{\dot{U}_s} = \frac{2}{4 - \omega^2 + j6\omega} \text{ (转移导纳)}$$

$$\frac{\dot{U}_L}{\dot{U}_s} = \frac{j2\omega}{4 - \omega^2 + j6\omega} \text{ (电压比)}$$

- 网络函数的阶数： $j\omega$ 的最高次方的次数。
本例为2阶

§ 11-2 RLC串联谐振电路

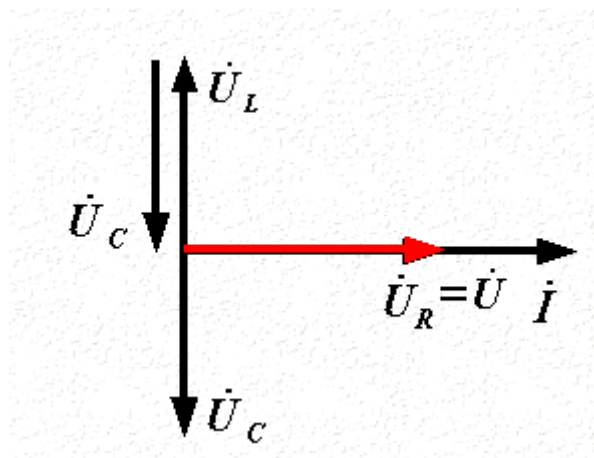
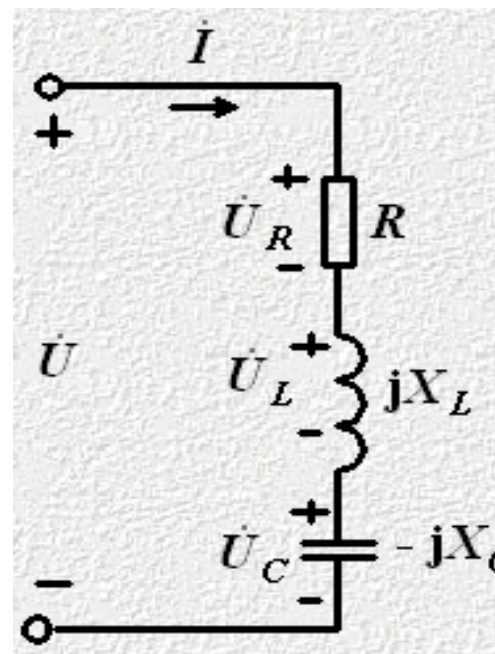
串联谐振

$$Z = \frac{\dot{U}}{\dot{I}} = R + j(X_L - X_C) = R + jX = |Z| \angle \varphi$$

1. 条件: $X_L - X_C = 0 \quad \omega L = \frac{1}{\omega C}$

2. 频率: $f = f_0 = \frac{1}{2\pi\sqrt{LC}}$

频率、角频率的关系为 $\omega = 2\pi f$



§ 11-2 RLC串联谐振电路

3. 特征:

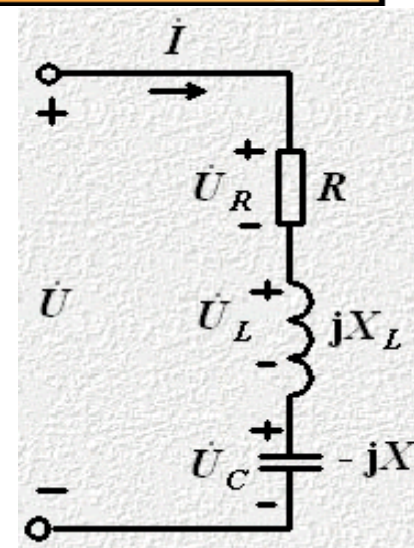
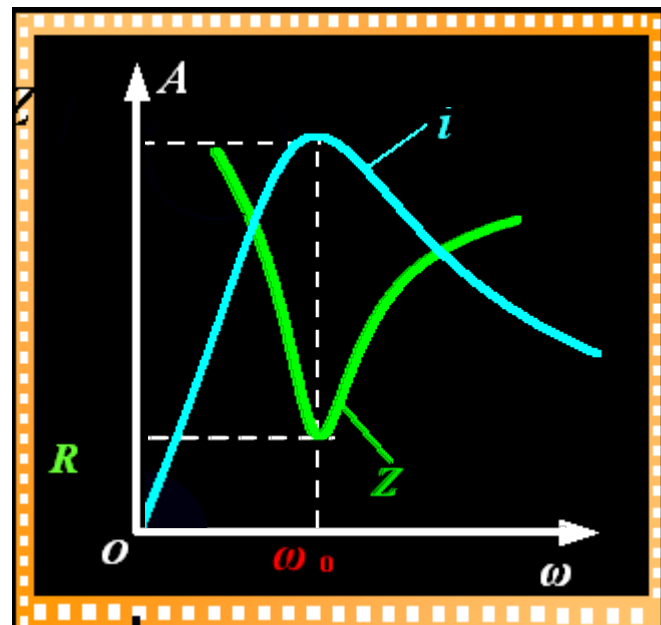
(1) $\dot{I}$ 、 $\dot{U}$ 同相 $\phi=0$

(2) $X_L = X_C$ $Z_{\min} = R$

$$I_{\max} = \frac{U}{R}$$

(3) $\dot{U}_L = -\dot{U}_C$ $\dot{U} = \dot{U}_R$ **电压谐振**

电阻两端电压值达到最大值，流过的电流达到最大值。



§ 11-2 RLC串联谐振电路

(4) 品质因数:

$$Q = \frac{U_L}{U_R}$$

Q 在此处不表示无功功率

$$= \frac{\omega_0 L}{R}$$

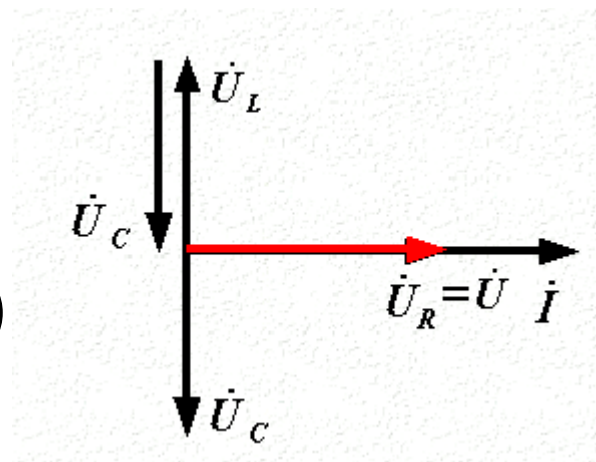
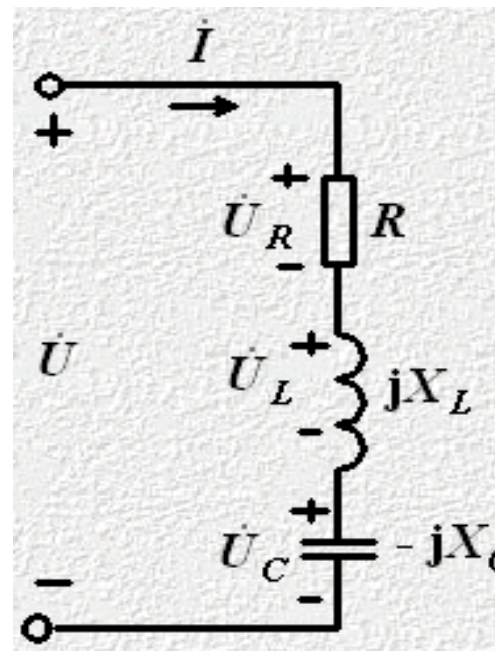
$$= \frac{1}{\omega_0 CR}$$

$$= \frac{1}{R} \sqrt{\frac{L}{C}}$$

$$U_L(j\omega_0) = U_C(j\omega_0) = QU_S(j\omega_0)$$

当 $X_L = X_C > R$ 时,

电容或电感上的电压 $U_L = U_C > U_S$



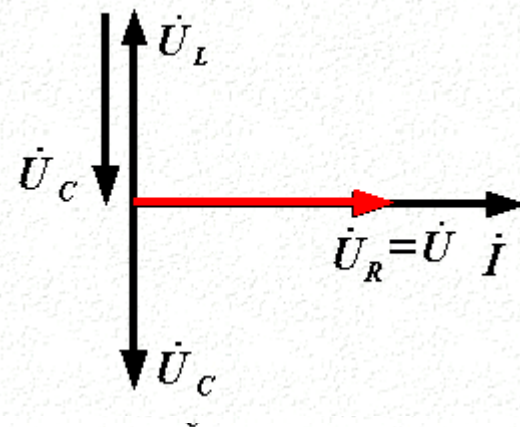
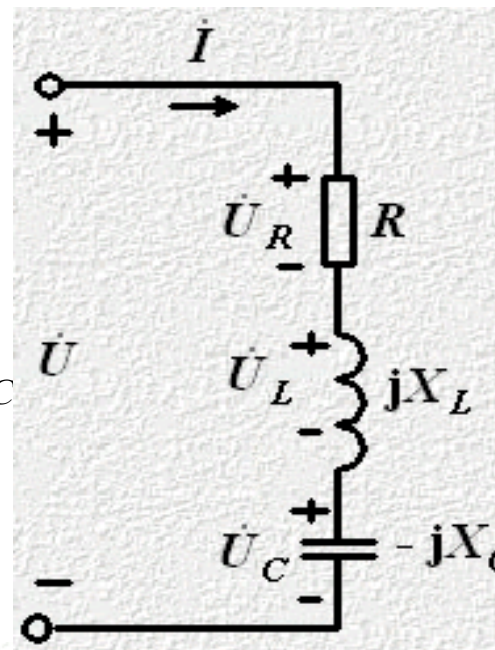
§ 11-2 RLC串联谐振电路

(5) 无功功率:

$$\begin{aligned} Q(j\omega_0) &= Q_L(j\omega_0) + Q_C(j\omega_0) \\ &= U_L I(j\omega_0) \sin \varphi_L + U_C I(j\omega_0) \sin \varphi_C \\ &= 0 \end{aligned}$$

此时电源 U_s 只发出有功功率供电阻 R 消耗

$$\bar{S}_S(j\omega_0) = RI^2(j\omega_0) = \frac{U_s^2(j\omega_0)}{R}$$



§ 11-2 RLC串联谐振电路

(6) 电感和电容中储存的电磁能:

$$\begin{aligned}W(j\omega_0) &= W_L(j\omega_0) + W_C(j\omega_0) \\&= \frac{1}{2} L I_m^2 \sin^2(\omega_0 t) + \frac{1}{2} C U_m^2 \cos^2(\omega_0 t) \\&= \frac{1}{2} L I_m^2 \sin^2(\omega_0 t) + \frac{1}{2} \frac{1}{\omega_0^2 L} ((\omega_0 L) I_m)^2 \cos^2(\omega_0 t) \\&= \frac{1}{2} L I_m^2 (j\omega_0) \\&= L I^2(j\omega_0) \\&= C Q^2 U_s^2(j\omega_0)\end{aligned}$$

式中 Q 为电路的品质因数。

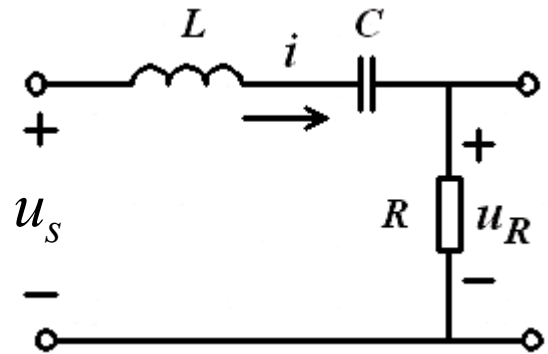
§ 11-2 RLC串联谐振电路

*RLC*串联电路的幅频特性和相频特性

$$N(j\omega) = \frac{\dot{U}_R}{\dot{U}_S} = \frac{R}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \quad \angle -\arctan \frac{\omega L - \frac{1}{\omega C}}{R}$$

$$A(\omega) = \frac{R}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

$$\varphi(\omega) = -\arctan \frac{\omega L - \frac{1}{\omega C}}{R}$$

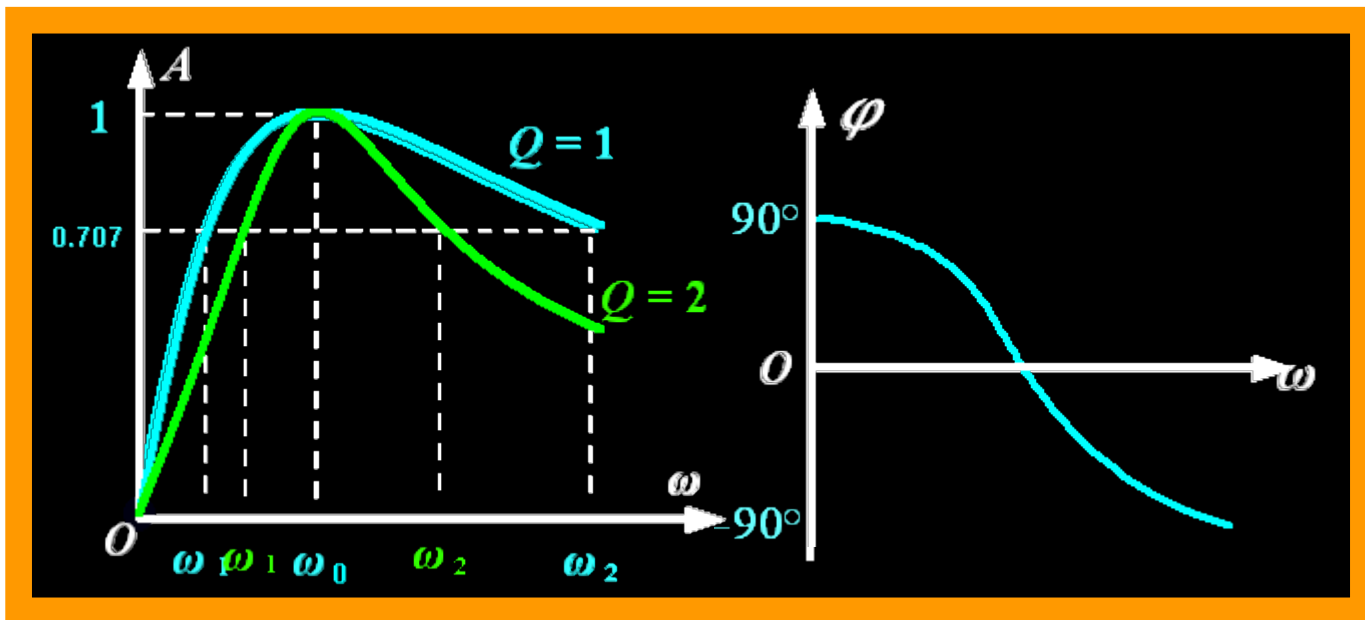
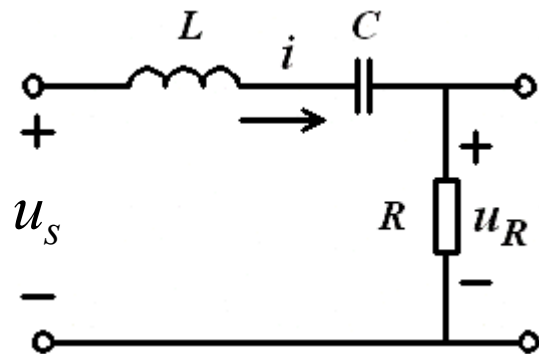


§ 11-2 RLC串联谐振电路

通频带:

$$BW = \omega_2 - \omega_1$$

$$BW = \frac{\omega_0}{Q}$$



§ 11-3 RLC并联谐振电路

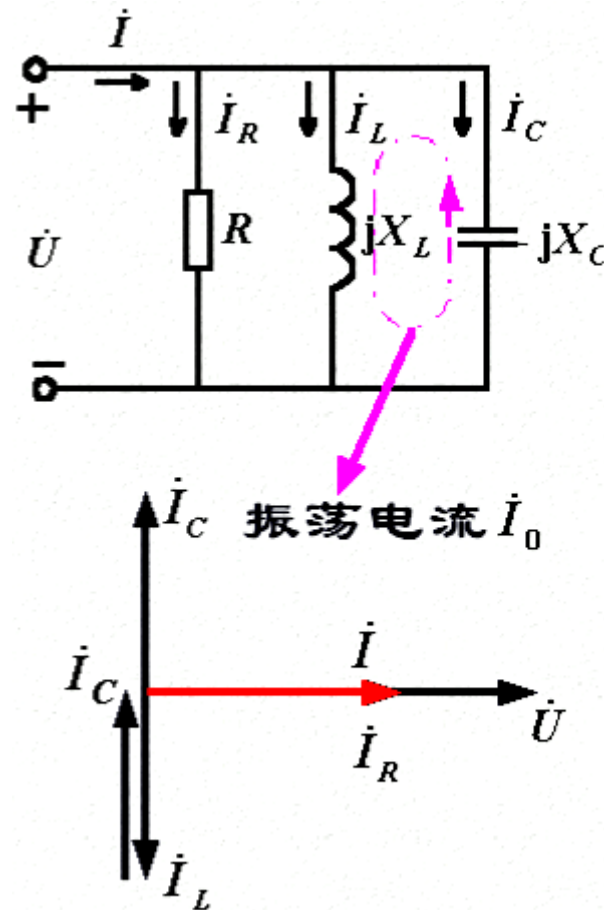
一、并联谐振

$$\dot{I} = \dot{I}_R + \dot{I}_L + \dot{I}_C$$

1. 条件: $\dot{I}_L = -\dot{I}_C$

$$\omega L = \frac{1}{\omega C}$$

2. 频率: $f = f_0 = \frac{1}{2\pi\sqrt{LC}}$



§ 11-3 RLC并联谐振电路

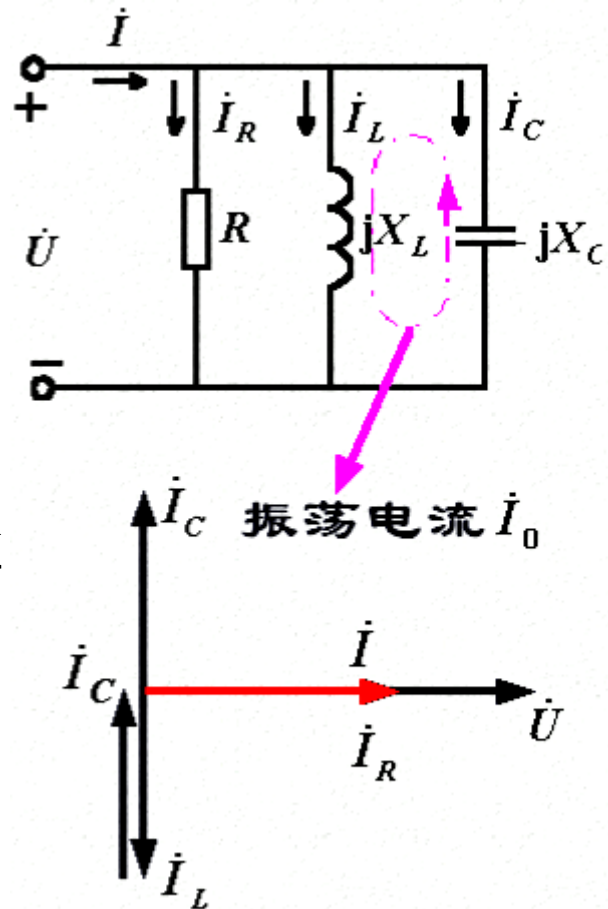
3. 特征：

(1) $\dot{U}$ 、 $\dot{I}$ 同相 $\phi=0$

(2) $Z_{\max} = R$ $I_{\min} = \frac{U}{R}$

(3) $\dot{I}_L + \dot{I}_C = 0$ $I = I_R$ 电流谐振

电源若为电流源，电阻中的电流即电流源的大小，此时端电压取得最大值。



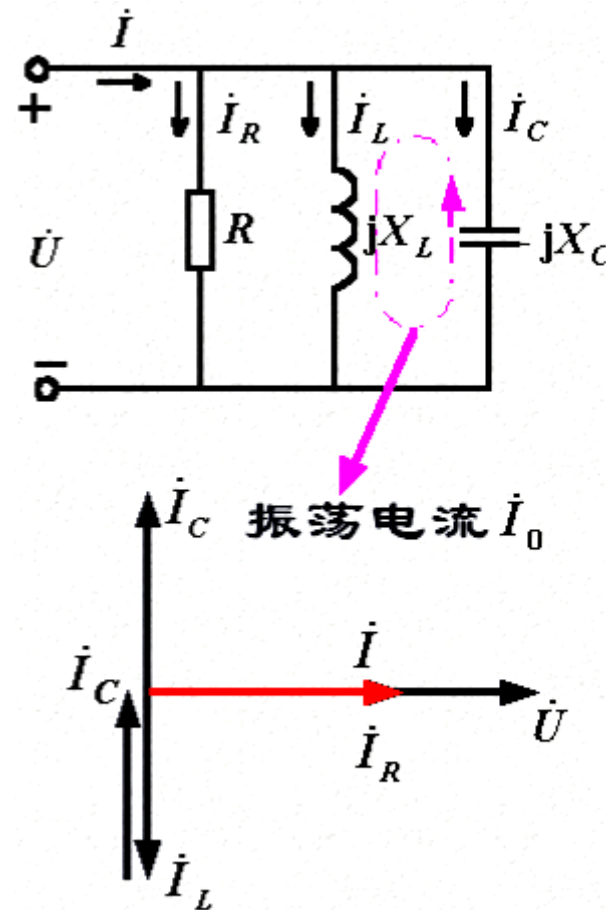
§ 11-3 RLC并联谐振电路

(4) 品质因数

$$Q = \frac{I_L}{I} = \frac{R}{X_L}$$

$$X_L = X_C < R \quad I_L = I_C > I$$

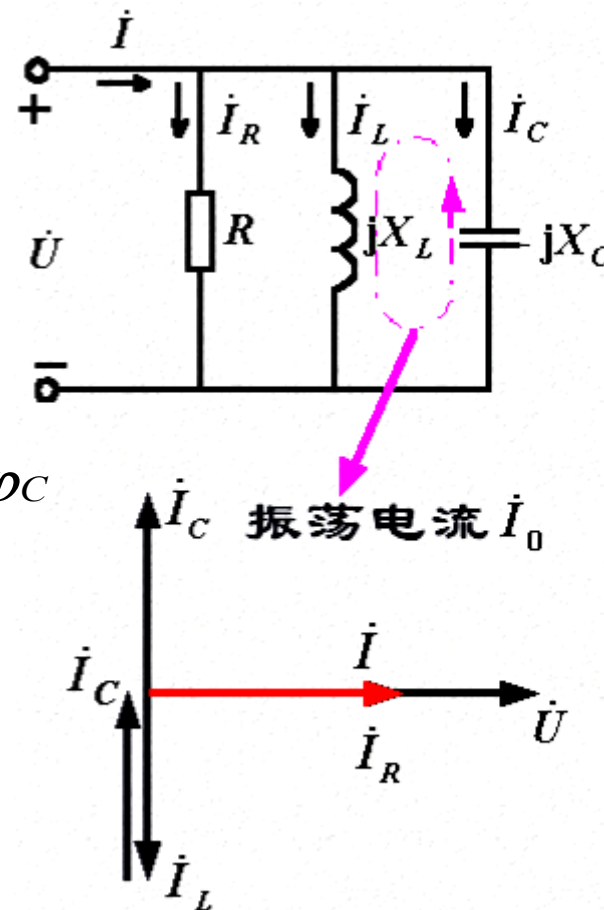
通频带: $BW = \frac{\omega_0}{Q}$



§ 11-3 RLC并联谐振电路

(5) 无功功率:

$$\begin{aligned} Q(j\omega_0) &= Q_L(j\omega_0) + Q_C(j\omega_0) \\ &= UI_L(j\omega_0) \sin \varphi_L + UI_C(j\omega_0) \sin \varphi_C \\ &= 0 \end{aligned}$$



§ 11-3 RLC并联谐振电路

(6) 电感和电容中储存的电磁能:

$$W(j\omega_0) = W_L(j\omega_0) + W_C(j\omega_0)$$

$$= \frac{1}{2} L I_m^2 \sin^2(\omega_0 t - \frac{\pi}{2}) + \frac{1}{2} C U_m^2 \sin^2(\omega_0 t)$$

$$= \frac{1}{2} L I_m^2 \cos^2(\omega_0 t) + \frac{1}{2} \frac{1}{\omega_0^2 L} ((\omega_0 L) I_m)^2 \sin^2(\omega_0 t)$$

$$= \frac{1}{2} L I_m^2 (j\omega_0)$$

$$= L I^2 (j\omega_0)$$

$$= L Q^2 I_s^2 (j\omega_0)$$

§ 11-3 RLC并联谐振电路

二、工程中采用的并联谐振电路：

$$Y = \frac{1}{(R + j\omega L)} + j\omega C$$

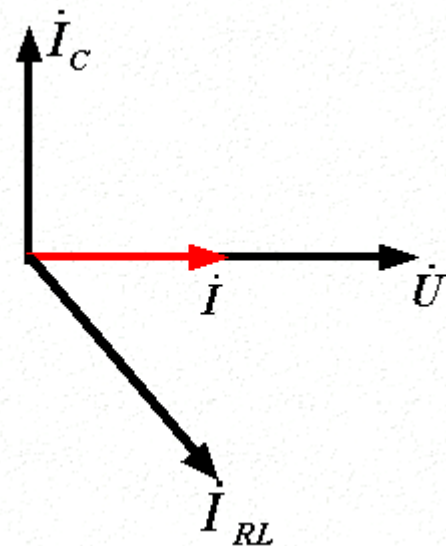
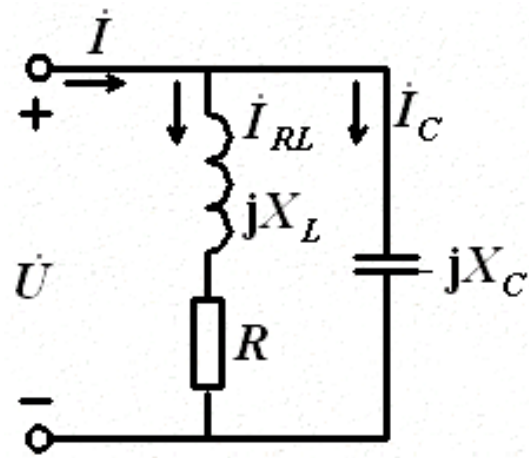
$$= \frac{R}{R^2 + (\omega L)^2} + j\left(\omega C - \frac{\omega L}{R^2 + (\omega L)^2}\right)$$

谐振时，导纳的虚部为0。

谐振频率： $\omega_0 = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}} \quad R < \sqrt{\frac{L}{C}}$

电源电压与电路中的电流同相 $\varphi=0$

若 $R > \sqrt{\frac{L}{C}}\omega_0$ 非实数，电路不发生谐振



§ 11-4 波特图

网络函数

$$H(j\omega) = \frac{\dot{R}_k(j\omega)}{\dot{E}_{sj}(j\omega)} = |H(\omega)| \angle \varphi(\omega) = |H(\omega)| e^{j\varphi(j\omega)}$$

两边取对数: $\ln H(j\omega) = \ln |H(\omega)| + j\varphi(j\omega)$

对数模用分贝(dB)表示 $H_{dB} = 20\lg |H(\omega)|$

横轴用 $\lg\omega$, 纵坐标用 H_{dB} , 得到幅频特性图;

横轴用 $\lg\omega$, 纵坐标用 $\Psi(j\omega)$, 得到相频特性图.

§ 11-5 滤波器简介

滤波电路——利用 L 、 C 对不同信号的频率呈现不同阻抗的性质，有目的地使某一段频率顺利通过或者得到有效抑制。

分类：低通、高通、带通、带阻。

§ 11-5 滤波器简介

1. 低通滤波器

$$N(j\omega) = \frac{\dot{U}_c}{\dot{U}_s}$$

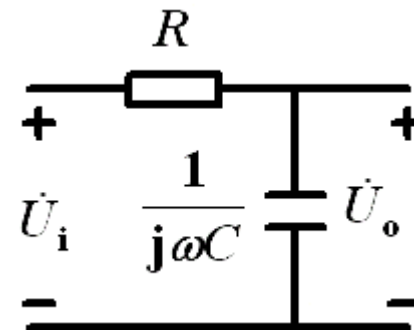
$$= \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}}$$

$$= \frac{1}{1 + jR\omega C}$$

$$= \frac{1}{\sqrt{1 + (R\omega C)^2}} \quad \text{—} \arctan(R\omega C)$$

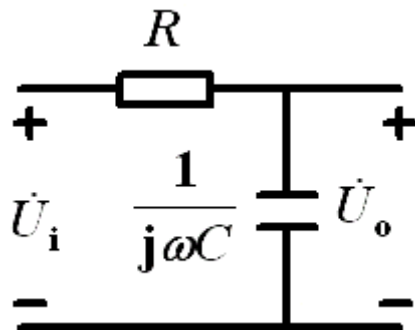
$$A(\omega) = \frac{1}{\sqrt{1 + (R\omega C)^2}}$$

$$\varphi(\omega) = -\arctan(R\omega C)$$



§ 11-5 滤波器简介

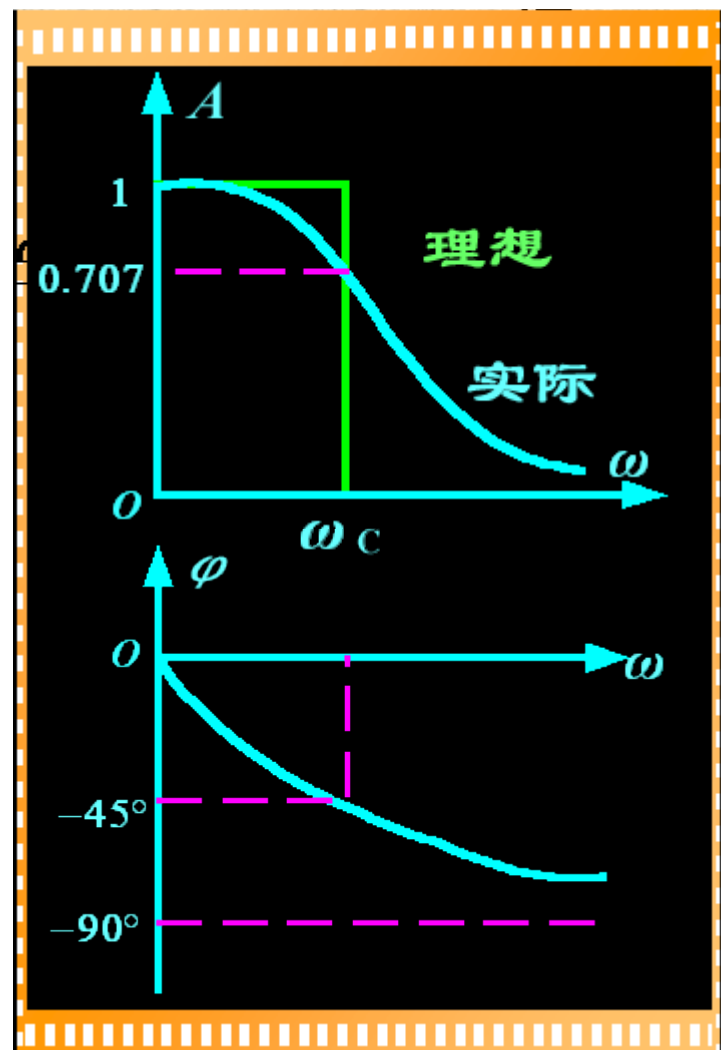
1. 低通滤波器



$$A(\omega) = \frac{1}{\sqrt{1 + (R\omega C)^2}}$$

$$\varphi(\omega) = -\arctan(R\omega C)$$

$$A(\omega) = \frac{1}{\sqrt{2}} \Rightarrow \omega_c = \frac{1}{RC}$$



§ 11-5 滤波器简介

2. 高通滤波器

$$N(j\omega) = \frac{\dot{U}_R}{\dot{U}_S}$$

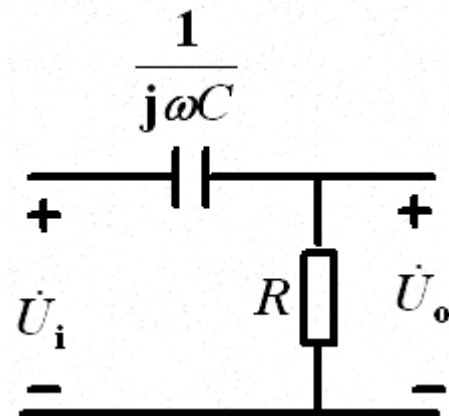
$$= \frac{R}{R + \frac{1}{j\omega C}}$$

$$= \frac{1}{1 + \frac{1}{jR\omega C}}$$

$$= \frac{1}{\sqrt{1 + \left(\frac{1}{R\omega C}\right)^2}} \angle \arctan\left(\frac{1}{R\omega C}\right)$$

$$A(\omega) = \frac{1}{\sqrt{1 + \left(\frac{1}{R\omega C}\right)^2}}$$

$$\phi(\omega) = \arctan\left(\frac{1}{R\omega C}\right)$$



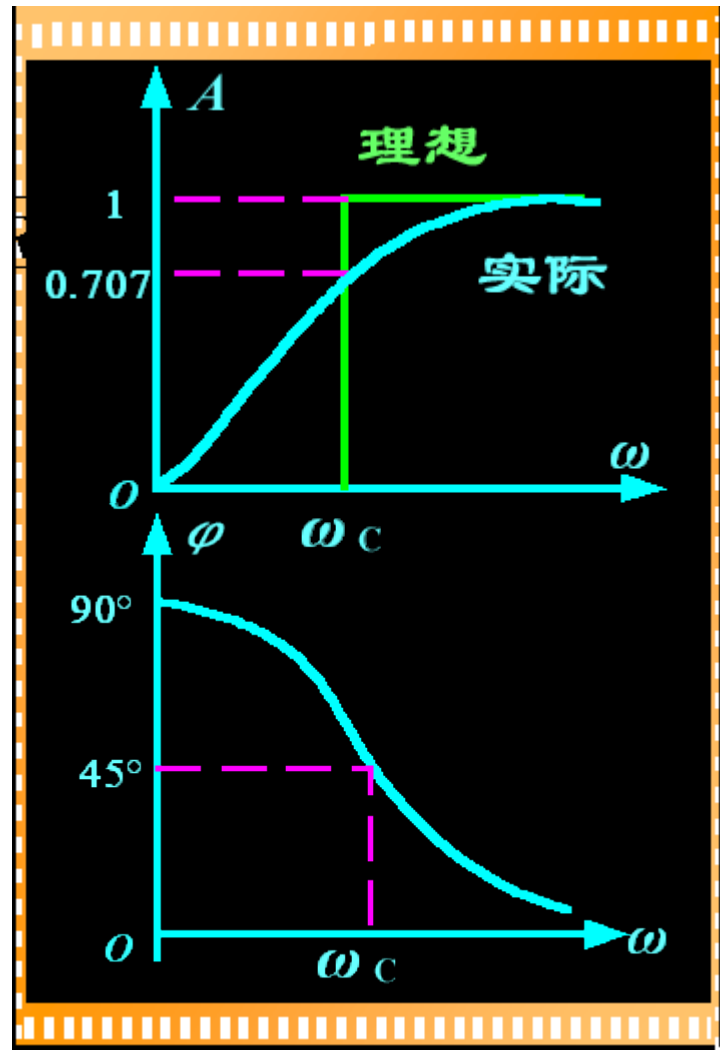
§ 11-5 滤波器简介

2. 高通滤波器

$$A(\omega) = \frac{1}{\sqrt{1 + \left(\frac{1}{R\omega C}\right)^2}}$$

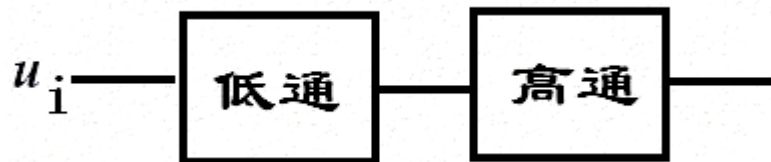
$$\phi(\omega) = \arctan\left(\frac{1}{R\omega C}\right)$$

$$A(\omega) = \frac{1}{\sqrt{2}} \Rightarrow \omega_c = \frac{1}{RC}$$



§ 11-5 滤波器简介

3. 带通滤波器



4. 带阻滤波器

